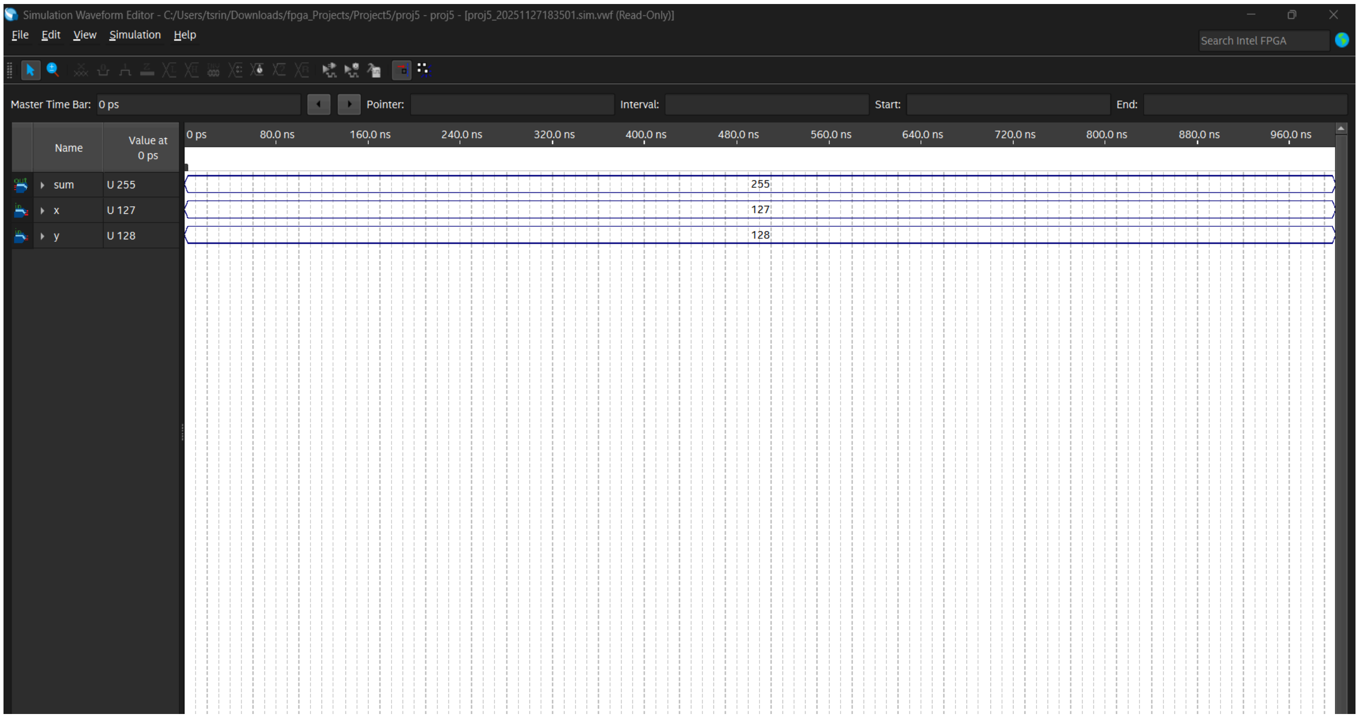
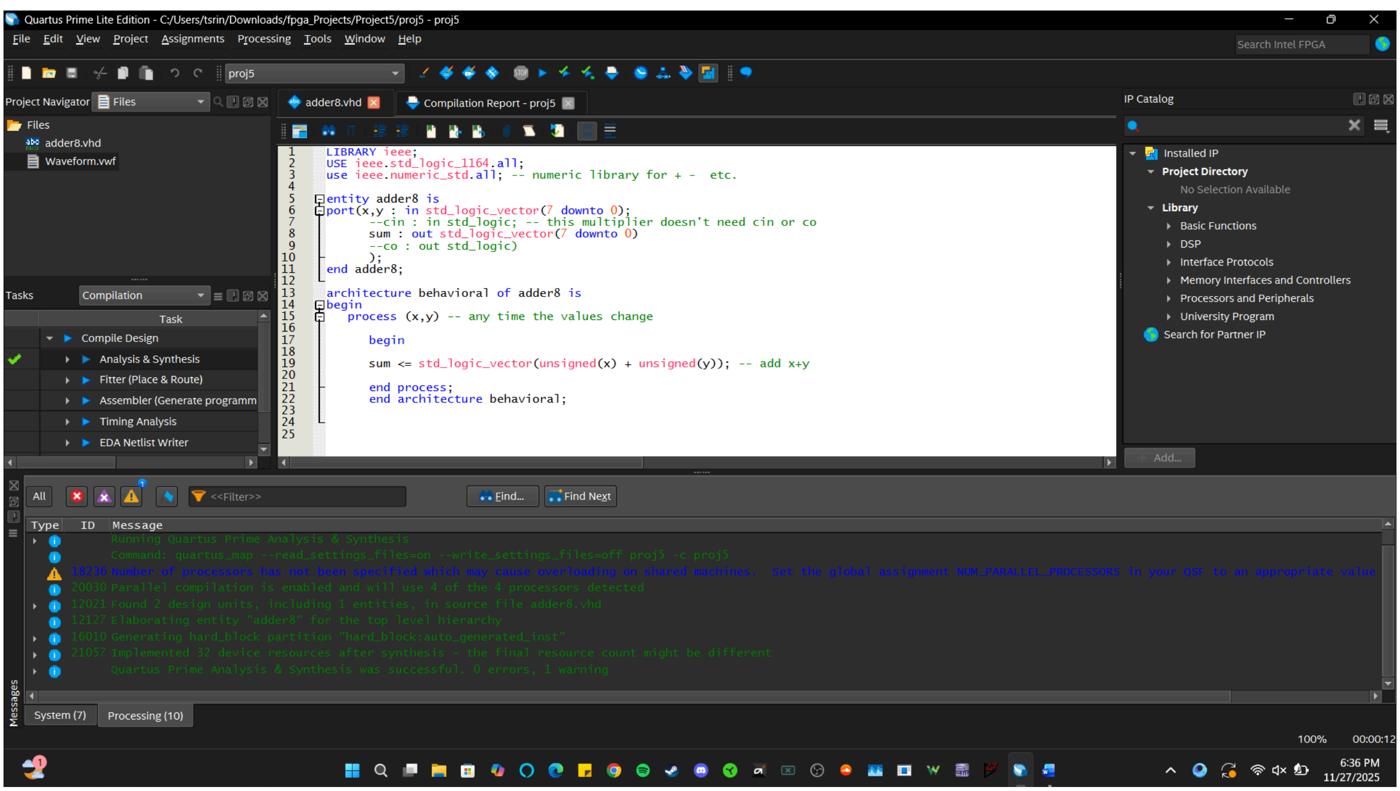
Name:

I pledge on my honor that I have not given or received any unauthorized assistance on this assessment _____________Sachin Srinivasan______________________________

PART I: Adder Design

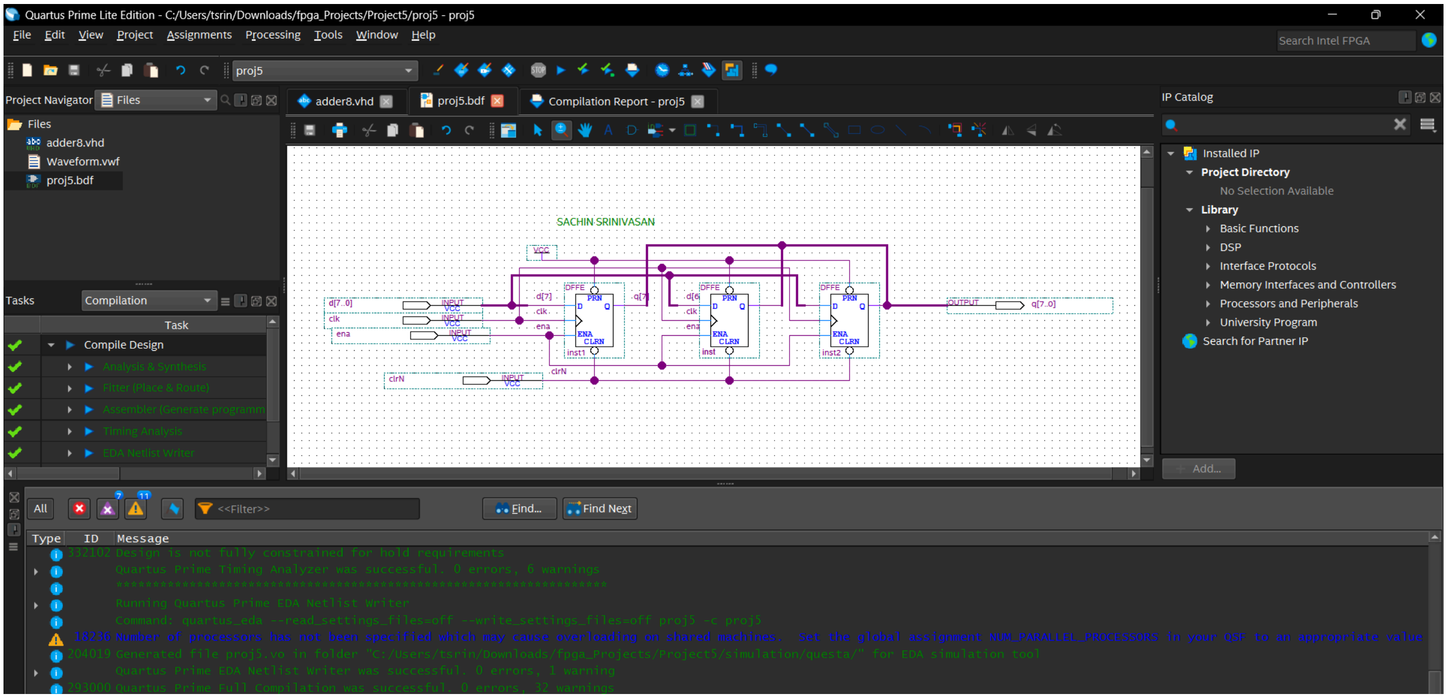
1. Verification of working 8-bit adder:





PART II: Accumulator Design

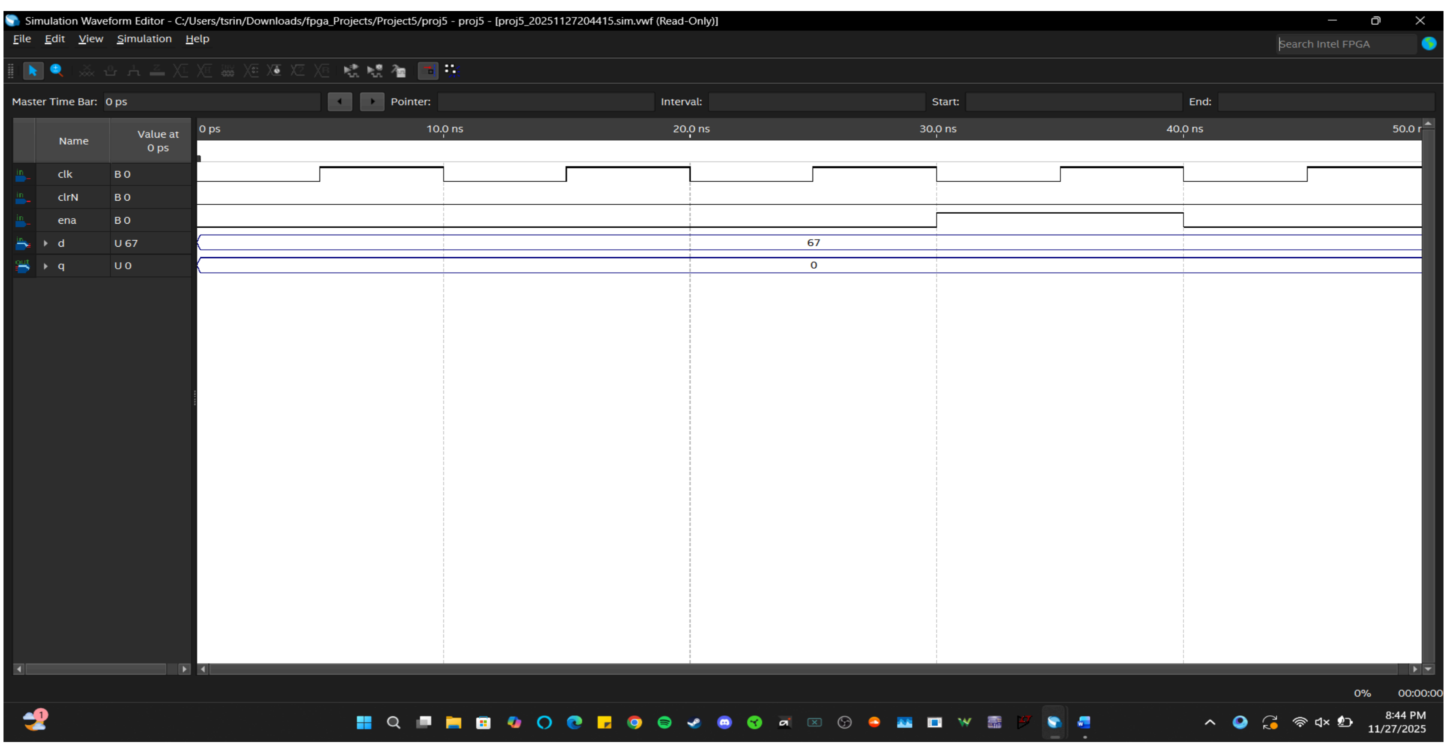
1. Screenshot of completed block diagram for 8-bit accumulator.



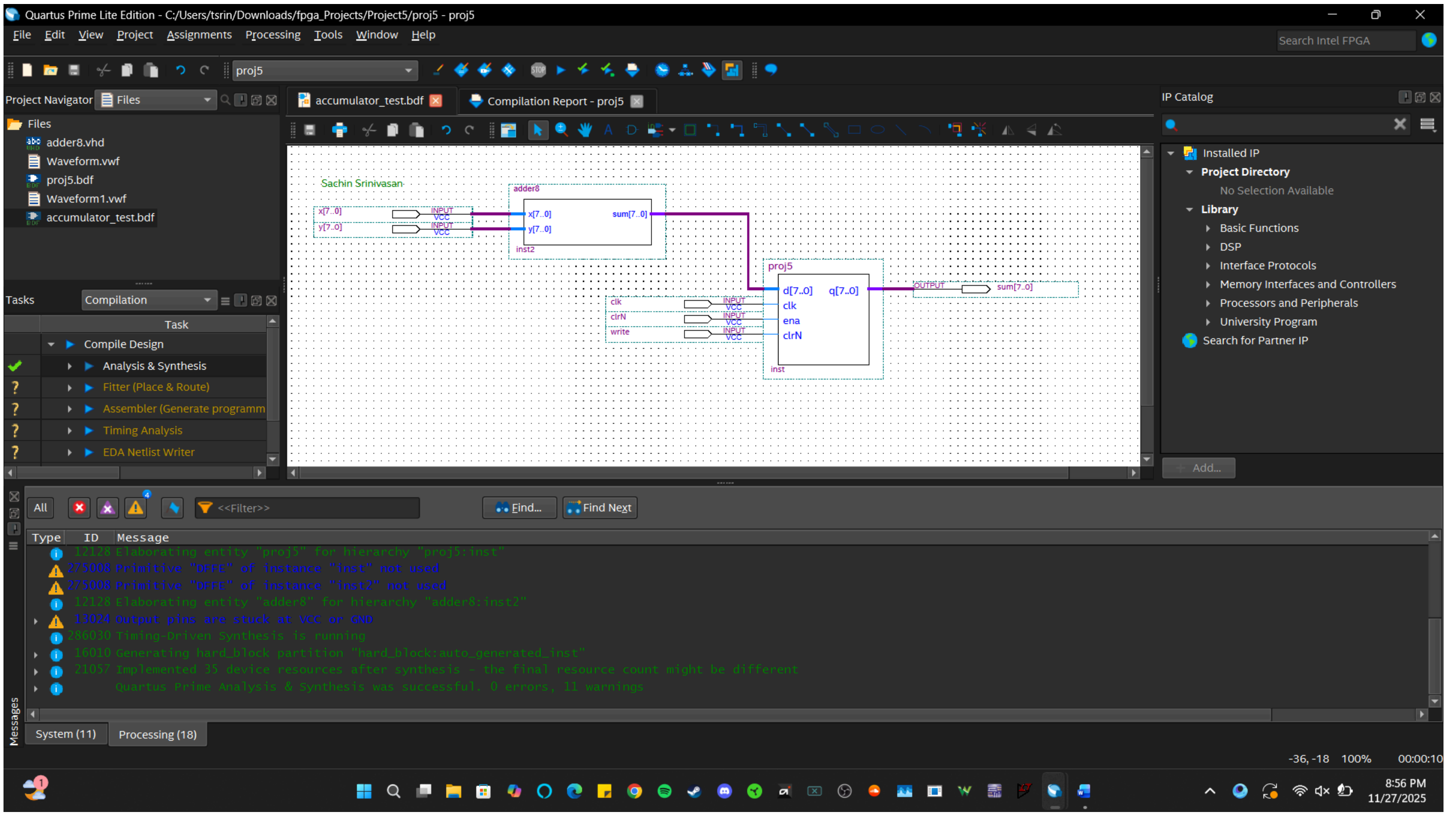
2. Question: Why are the PRN pins on the dffe's set to vcc?

The PRN pins on the dffe are set to vcc is so the flip-flops don't try to go back to 1. We would only want the accumulator to change on clr or during normal operation.

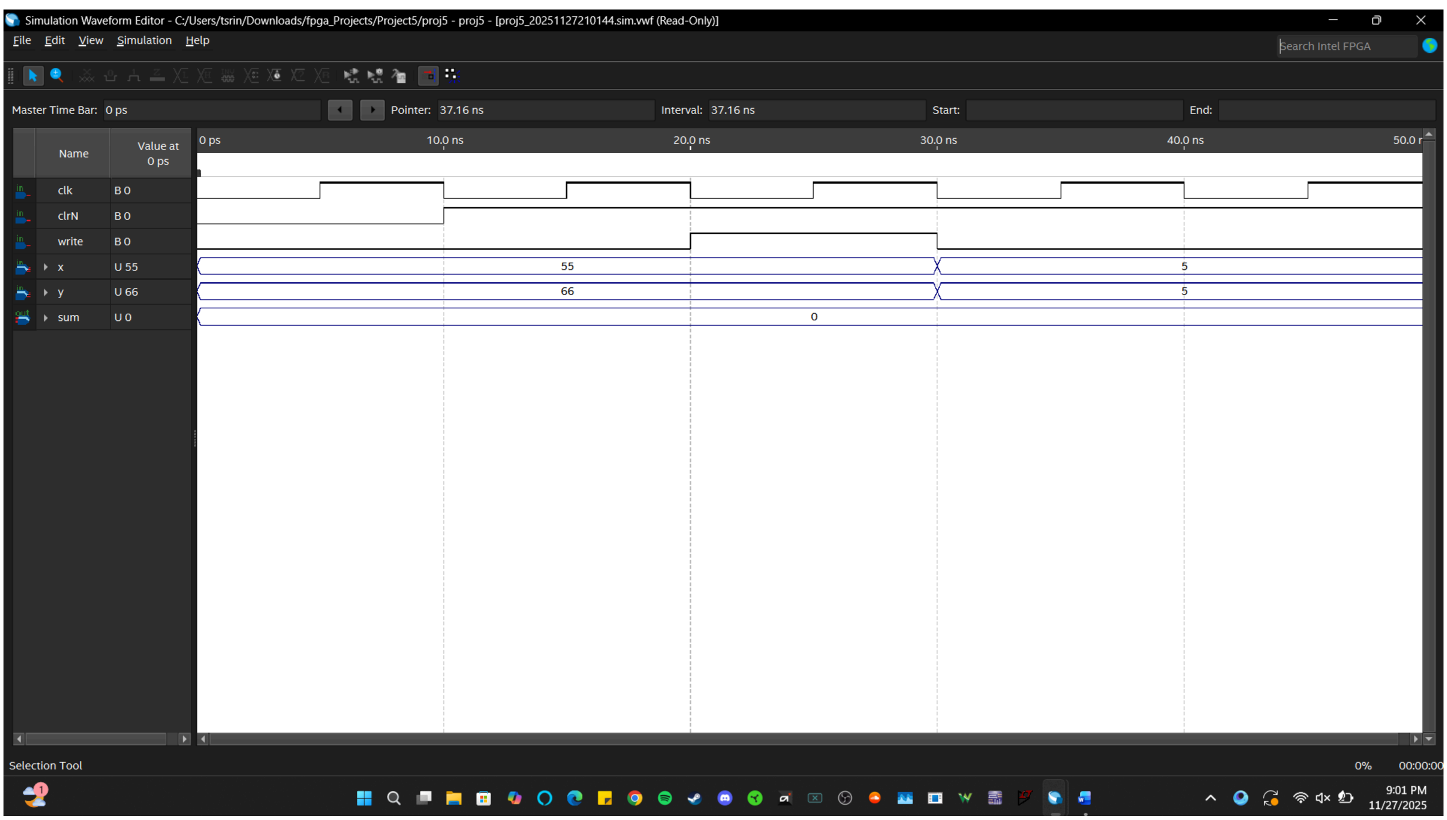
3. Screenshot of completed accumulator simulation.



5. Screenshot of completed adder / accumulator block diagram.



6. Screenshot of completed adder / accumulator simulation.



PART III: Counter Design

1. **State/Transition Table:**

PS ($Q_3Q_2Q_1Q_0$)	NS ($Q_3^+Q_2^+Q_1^+Q_0^+$)	DONE
0000	0000	1
0001	0000	0
0010	0001	0
0011	0010	0
0100	0011	0
0101	0100	0
0110	0101	0
0111	0110	0
1000	0111	0
1001	1000	0
1010	1001	0
1011	1010	0
1100	1011	0
1101	1100	0
1110	1101	0
1111	1110	0

2. **Minimal SOP for each next state equation and DONE:**

Q_3^+ = Q3 (Q2 + Q0 + Q1)

DONE = Q3’ Q2’ Q1’ Q0’

Q_2^+ = Q3Q2’Q1’Q0’ + Q2(Q0 + Q1)

Q_1^+ = Q1’Q0’(Q2+Q3)+Q1Q0

Q_0^+ = Q0’(Q1+Q2+Q3)

Work (including KMAPs) attached here.

KMAP FOR Q0+	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	0	1
10	1	0	0	1

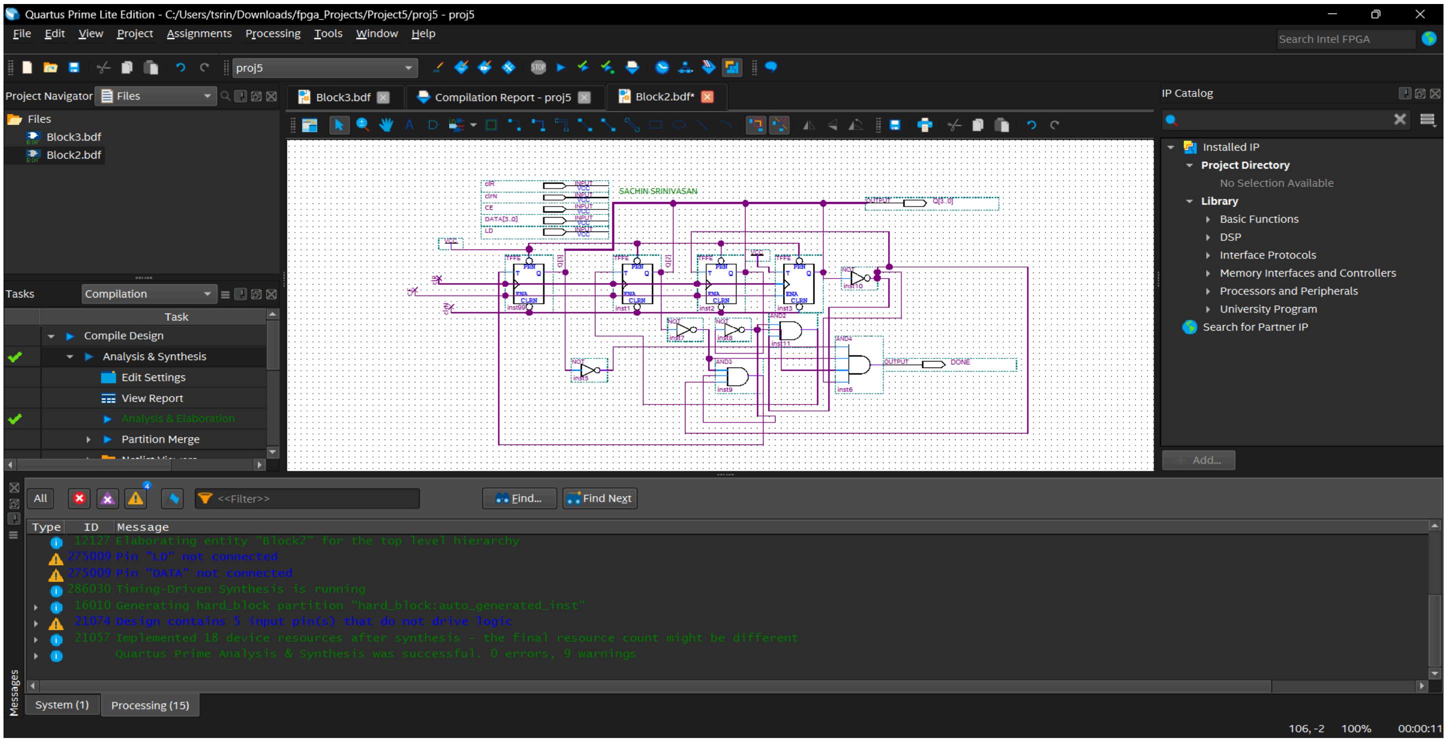
KMAP FOR Q1+	00	01	11	10
00	0	0	1	0
01	1	0	1	0
11	1	0	1	0
10	1	0	1	0

KMAP FOR Q2+	00	01	11	10
00	0	0	0	0
01	0	1	1	0
11	1	0	1	0
10	1	0	0	0

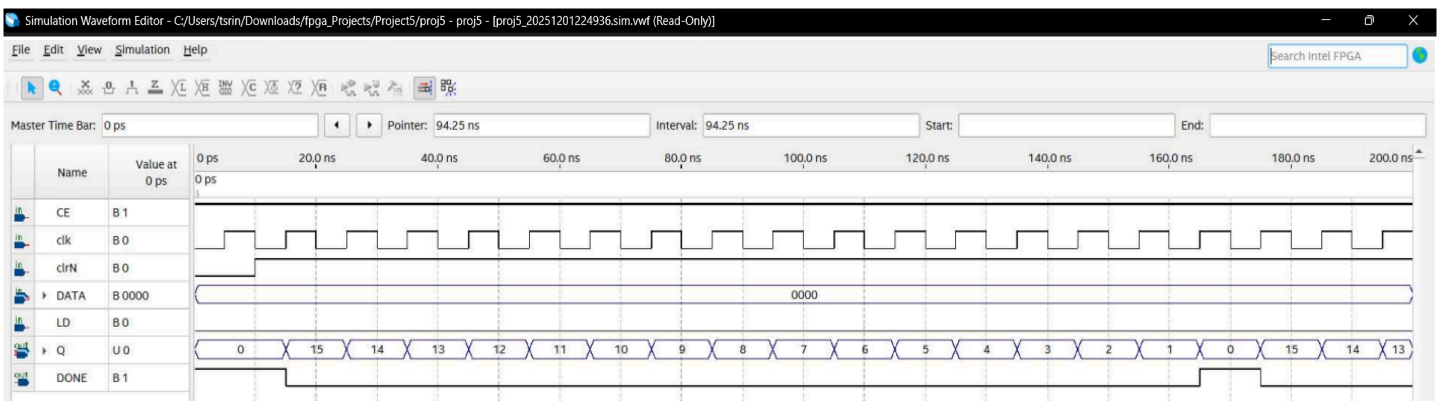
KMAP FOR Q3+	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	1	1	0

KMAP FOR Done	00	01	11	10
00	1	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

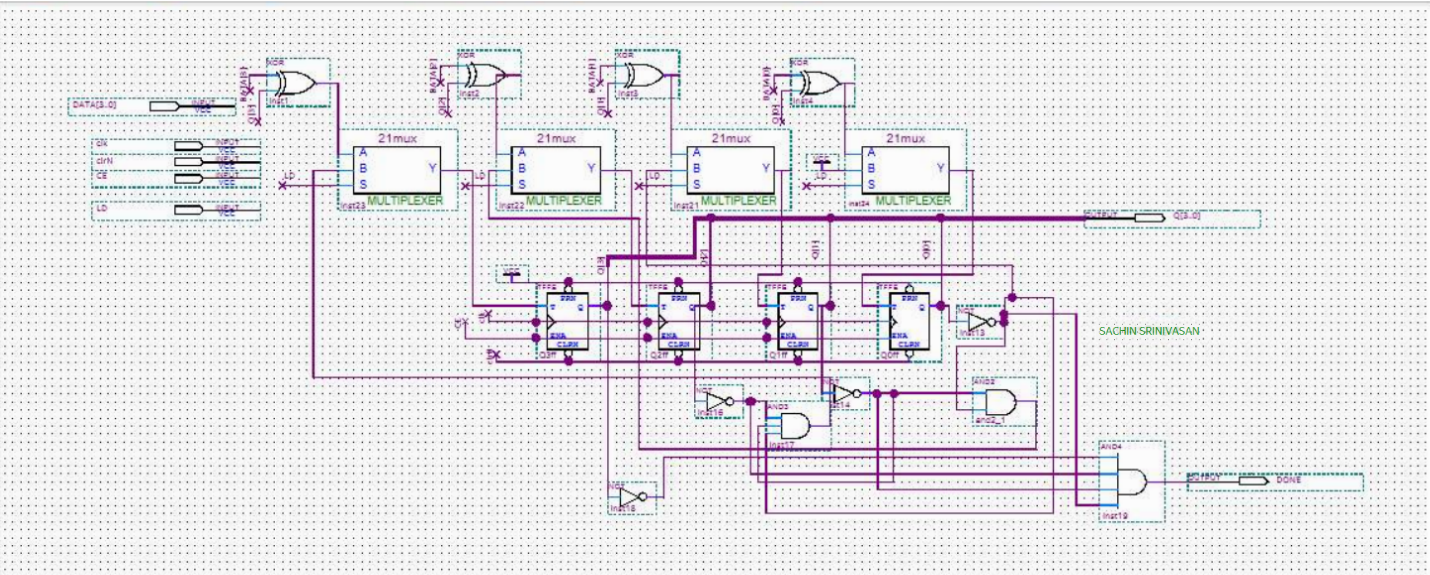
5. **Screenshot of the completed TFF binary DOWN counter**



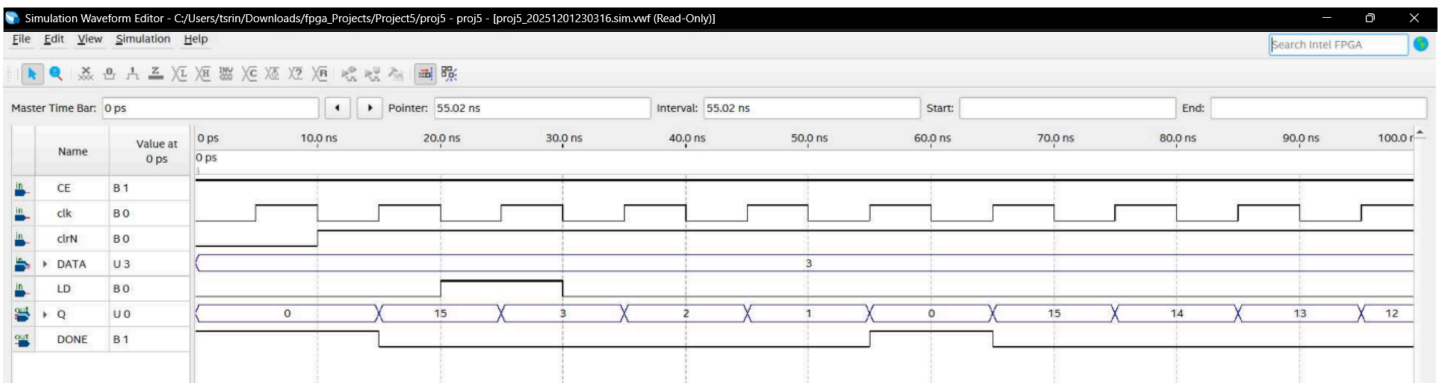
6. Screenshot of the completed DOWN counter simulation



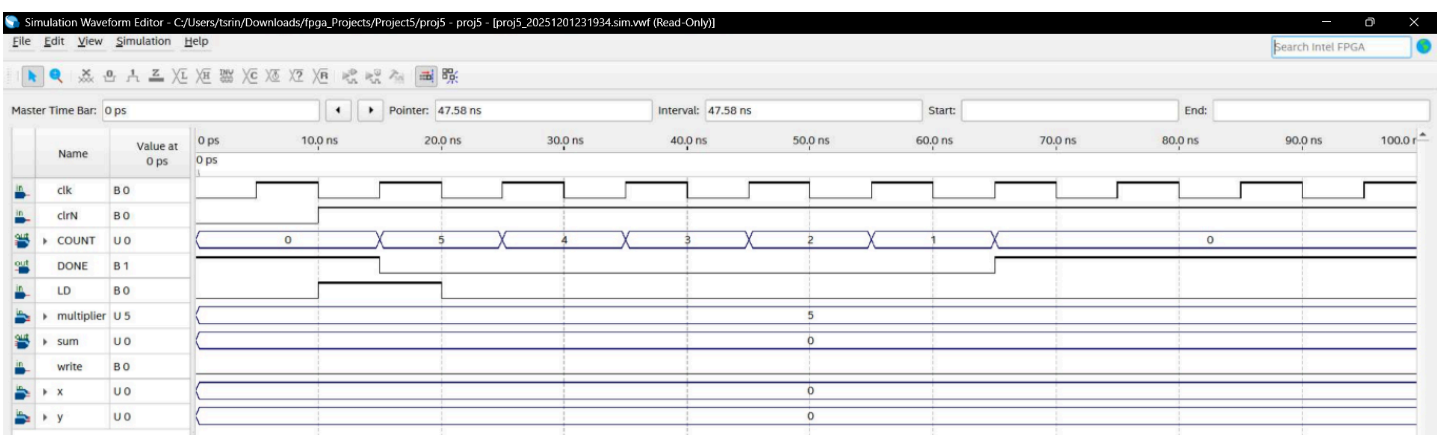
8. Screenshot of the completed TFF binary DOWN counter with synchronous load



15. Screenshot of the completed synchronous load counter simulation

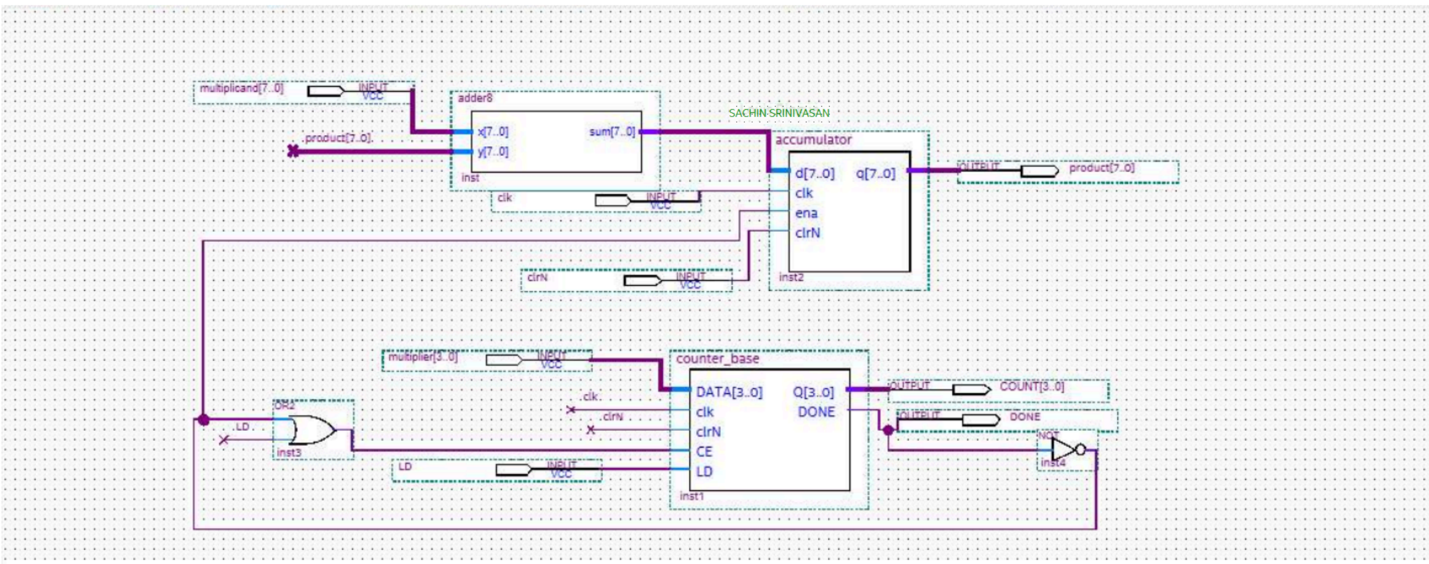


16. Screenshot of the completed counter simulation

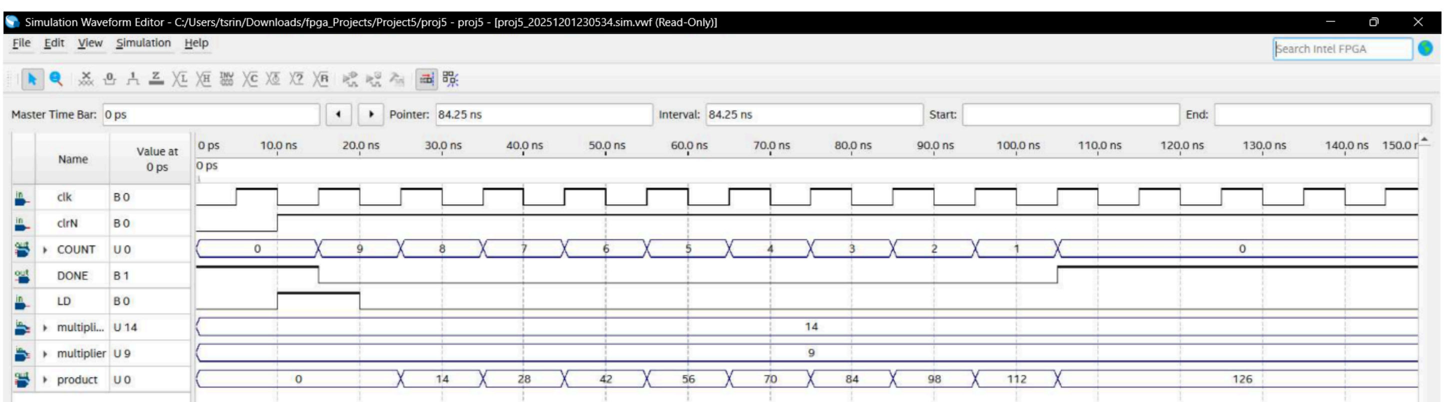


PART IV. Completing the Multiplier

1. Screenshot of completed multiplier circuit diagram



2. Screenshot of completed multiplier circuit simulation



Questions:

- Why is the accumulator enable set to NOT DONE?
  When DONE becomes 1, the operation is finished, so using NOT DONE stops further updates.

- If the multiplicand is limited to 4-bits, what are the largest numbers you can multiply and what is the largest product this circuit will generate (assume unsigned numbers)?
  The largest multiplication is 15 × 15 = 225.

- What would happen if you tried to multiply 250 * 3?
  250 can't fit in 4 bits, so the circuit would only see the lower 4 bits and compute 10 × 3 instead of 250 × 3.

- How do you think a circuit like this could prevent the multiplicand from being more than 4-bits ?
  By adding a small check that detects if any upper bits are 1 and blocking the load if the value exceeds 4 bits.

- Do you think this is an efficient way to build a multiplier? Why or why not?

It depends because it uses repeated addition, which takes multiple cycles instead of calculating the product in one step.